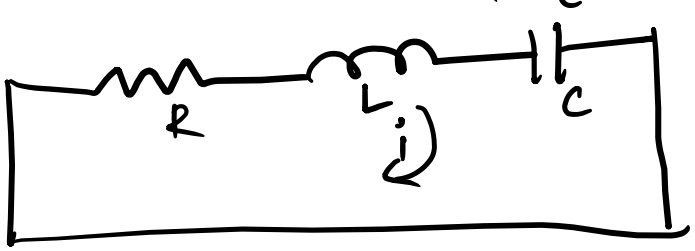
RC $v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\, e^{-t/\tau}$

RL: $i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]\, e^{-t/\tau}$

RLC circuits



$i = i_R = i_L = i_C$

$$v_R + v_L + v_C = 0$$

In terms of i_L

$$R \cdot i_L + L \frac{di_L}{dt} + \frac{1}{C} \int_0^t i_L \, dt = 0$$

$$\frac{d^2 i_L}{dt^2} + \frac{R}{L} \frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

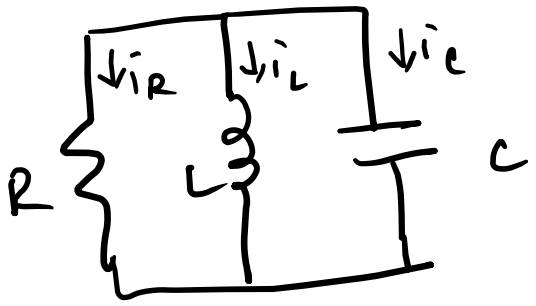
In terms of i_C

$$R i_C + L \frac{di_C}{dt} + v_C = 0$$

$$RC \frac{dv_C}{dt} + LC \frac{d^2 v_C}{dt^2} + v_C = 0$$

$$\frac{d^2 v_C}{dt^2} + \frac{R}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = 0$$

Parallel RLC



$$i_R + i_L + i_C = 0$$

$$\frac{v_L}{R} + i_L + C\frac{dv_L}{dt} = 0$$

$$\frac{L}{R}\frac{di_L}{dt} + i_L + LC\frac{d^2 i_L}{dt^2} = 0 \quad \left(\text{Replacing } v_L \text{ with } L\frac{di_L}{dt}\right)$$

$$\frac{d^2 i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

$$\frac{d^2 v_C}{dt^2} + \frac{1}{RC}\frac{dv_C}{dt} + \frac{1}{LC} v_C = 0$$

Standard 2nd order diff eq.

$$\frac{d^2 x}{dt^2} + b\frac{dx}{dt} + cx = 0$$

Characteristic equation.

$$s^2 + bs + c = 0$$

$$s_{12} = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

If $b^2 - 4c > 0$

$$x(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

If $b^2 - 4c < 0$

$$s_1 = -\sigma + j\omega_d$$

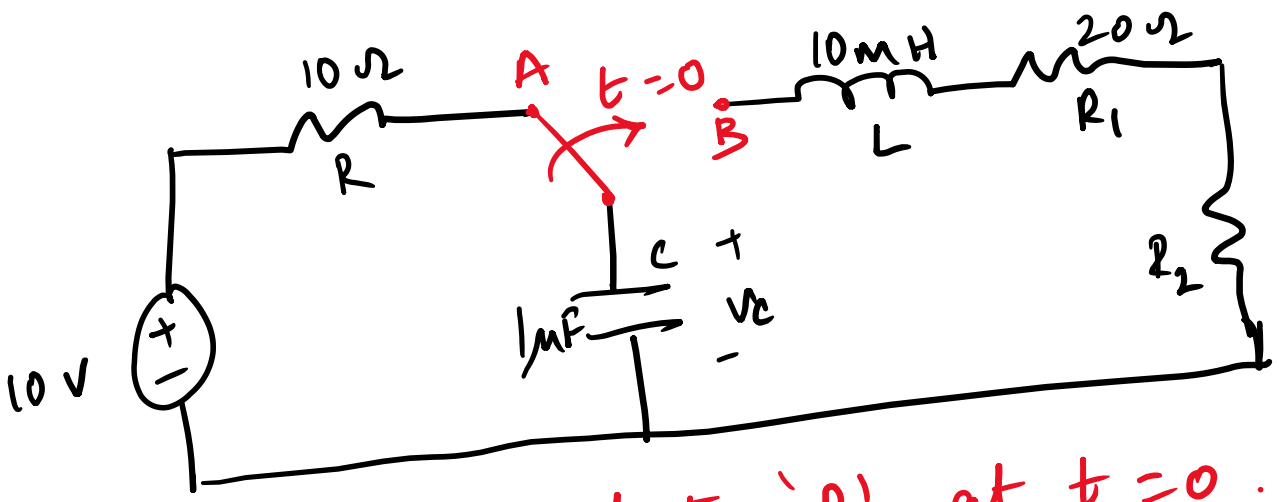
$$s_2 = -\sigma - j\omega_d$$

$$x(t) = e^{-\sigma t} [A\cos(\omega_d t) + B\sin(\omega_d t)]$$

If $b^2 = 4c$

$$x(t) = (k_1 + k_2 t) e^{s_1 t}$$

- Over damped
- Underdamped
- Critically damped.



Switch is flipped to 'B' at $t=0$.

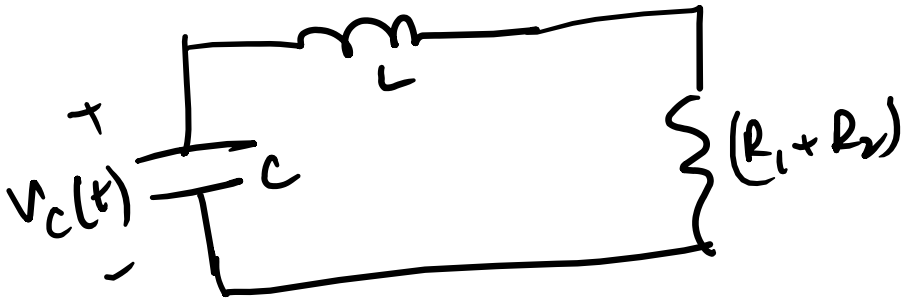
Find $V_c(t)$

a) $R_2 = 405\ \Omega$

b) $R_2 = 0\ \Omega$

c) $R_2 = 180\ \Omega$

For $t \geq 0$



$$\frac{d^2V_c}{dt^2} + \frac{R}{L}\frac{dV_c}{dt} + \frac{1}{LC}V_c = 0$$

$$\frac{d^2V_c}{dt^2} + \frac{(20+R_2)}{0.01}\frac{dV_c}{dt} + \frac{1}{10^{-8}}V_c = 0$$

$$s^2 + 100(20+R_2)s + 10^8 = 0$$

$$s_{1,2} = \frac{-100(20+R_2) \pm \sqrt{(100(20+R_2))^2 - 4\times10^8}}{2}$$

(a) $R_2 = 40\,\Omega$ $\quad b^2 - 4c > 0$

$s_1 = -2500$

$s_2 = -40000$

$$v_C(t) = k_1 e^{-2500t} + k_2 e^{-40000t}$$

$$v_C(0^-) = v_C(0) = v_C(0^+) = 10\,V$$

$$k_1 + k_2 = 10 \quad - (1)$$

$$i_L(0^-) = i_L(0) = i_L(0^+) = 0$$

$$-C \left.\frac{dv_C(t)}{dt}\right|_{t=0} = 0$$

$$-2500 k_1 e^{-2500t} - 40000 k_2 e^{-40000t} \Big|_{t=0} = 0$$

$$-2500 k_1 - 40000 k_2 = 0 \quad - (2)$$

$$k_1 = 10.667 \qquad k_2 = -0.667$$

$$v_C(t) = 10.667 e^{-2500t} - 0.667 e^{-40000t}$$

Overdamped

(b) $R_2 = 0$

$s_1 = -1000 + j\,9950$

$s_2 = -1000 - j\,9950$

$\rightarrow v_C(t) = e^{-1000t}\left[A\cos(9950t) + B\sin(9950t)\right]$

$v_C(0) = 10V \rightarrow A = 10$

$\left.\frac{dv_C(t)}{dt}\right|_{t=0} = 0 \qquad -1000A + 9950B = 0$

$B = 1.005$

$v_C(t) = e^{-1000t}\left[10\cos 9950t + 1.005\sin 9950t\right]$

(c) $R = 180 \qquad b^2 - 4c = 0 \qquad s_1 = -10^4$

$v_C(t) = (k_1 + k_2 t)e^{s_1 t}$

$v_C(t) = (k_1 + k_2 t)e^{-10^4 t}$

$v_C(0) = 10V \rightarrow k_1 = 10$

$\left.\frac{dv_C}{dt}\right|_{t=0} = 0 \rightarrow -10^4 k_1 + k_2 = 0$

$k_2 = 10^5$

$v_C(t) = (10 + 10^5 t)e^{-10^4 t}$